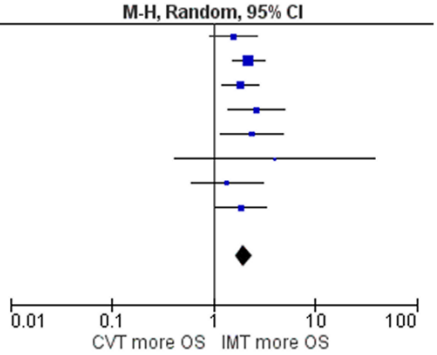
(a)

Study or Subgroup	IMT Events	IMT Total	CVT Events	CVT Total	Weight	Odds Ratio M-H, Random, 95% CI	Year
Maury 2016	64	105	52	104	12.7%	1.56 [0.90, 2.70]	2016
Gokbuget 2016	53	189	167	1115	30.0%	2.21 [1.55, 3.16]	2016
Kantarjian 2017	146	271	52	134	21.6%	1.84 [1.21, 2.81]	2017
Kantarjian 2019	37	164	16	162	9.6%	2.66 [1.41, 5.01]	2019
Jabbour 2019	31	58	25	77	7.8%	2.39 [1.18, 4.82]	2019
Alduailej 2020	7	8	14	22	0.7%	4.00 [0.41, 38.65]	2020
Rambaldi 2020	18	45	18	55	5.7%	1.37 [0.60, 3.11]	2020
Gokbuget 2020	49	73	95	182	11.9%	1.87 [1.06, 3.30]	2020
Total (95% CI)		**913**		**1851**	**100.0%**	**2.00 [1.64, 2.43]**	
Total events	405		439				

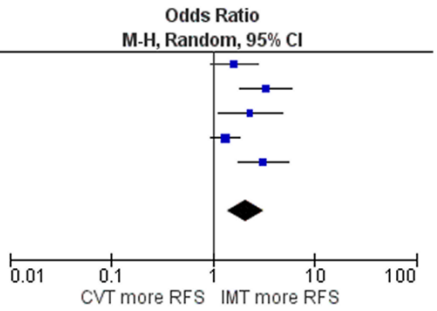
Heterogeneity: Tau² = 0.00; Chi² = 3.49, df = 7 (P = 0.84); I² = 0%
Test for overall effect: Z = 6.91 (P < 0.00001)



(b)

Study or Subgroup	IMT Events	IMT Total	CVT Events	CVT Total	Weight	Odds Ratio M-H, Random, 95% CI	Year
Maury 2016	58	105	45	104	19.9%	1.62 [0.94, 2.79]	2016
Kantarjian 2017	84	271	16	134	19.0%	3.31 [1.85, 5.93]	2017
Jabbour 2019	28	58	22	77	15.9%	2.33 [1.14, 4.76]	2019
Marks 2019	141	289	121	288	25.8%	1.31 [0.95, 1.83]	2019
Gokbuget 2020	46	73	64	182	19.4%	3.14 [1.79, 5.52]	2020
Total (95% CI)		**796**		**785**	**100.0%**	**2.12 [1.41, 3.19]**	
Total events	357		268				

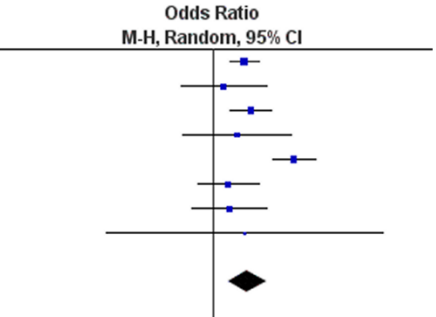
Heterogeneity: Tau² = 0.14; Chi² = 12.02, df = 4 (P = 0.02); I² = 67%
Test for overall effect: Z = 3.61 (P = 0.0003)



(c)

Study or Subgroup	IMT Events	IMT Total	CVT Events	CVT Total	Weight	Odds Ratio M-H, Random, 95% CI	Year
Gokbuget 2016	81	189	186	694	19.5%	2.05 [1.47, 2.86]	2016
Maury 2016	97	105	94	104	10.3%	1.29 [0.49, 3.41]	2016
Kantarjian 2017	119	271	33	134	17.6%	2.40 [1.51, 3.80]	2017
Jabbour 2019	53	57	68	77	7.8%	1.75 [0.51, 6.01]	2019
Kantarjian 2019	121	164	50	162	17.3%	6.30 [3.89, 10.21]	2019
Marks 2019	274	289	267	288	14.1%	1.44 [0.73, 2.85]	2019
Rambaldi 2020	16	45	15	55	11.8%	1.47 [0.63, 3.45]	2020
Alduailej 2020	8	8	20	22	1.7%	2.07 [0.09, 47.89]	2020
Total (95% CI)		**1128**		**1536**	**100.0%**	**2.20 [1.44, 3.37]**	
Total events	769		733				

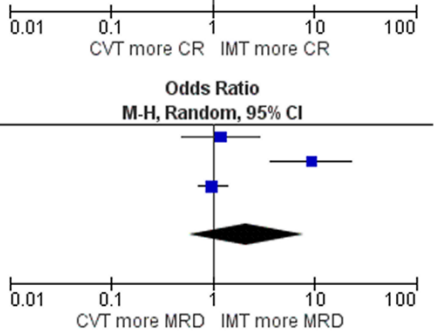
Heterogeneity: Tau² = 0.21; Chi² = 21.55, df = 7 (P = 0.003); I² = 68%
Test for overall effect: Z = 3.64 (P = 0.0003)



(d)

Study or Subgroup	IMT Events	IMT Total	CVT Events	CVT Total	Weight	Odds Ratio M-H, Random, 95% CI	Year
Maury 2016	32	49	22	36	31.8%	1.20 [0.49, 2.92]	2016
Kantarjian 2019	69	88	9	32	31.5%	9.28 [3.69, 23.36]	2019
Marks 2019	120	287	121	287	36.7%	0.99 [0.71, 1.37]	2019
Total (95% CI)		**424**		**355**	**100.0%**	**2.13 [0.59, 7.72]**	
Total events	221		152				

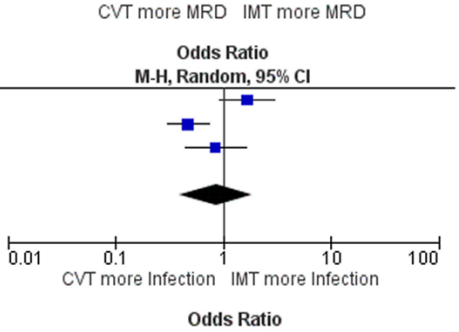
Heterogeneity: Tau² = 1.15; Chi² = 20.13, df = 2 (P < 0.0001); I² = 90%
Test for overall effect: Z = 1.15 (P = 0.25)



(e)

Study or Subgroup	IMT Events	IMT Total	CVT Events	CVT Total	Weight	Odds Ratio M-H, Random, 95% CI	Year
Maury 2016	40	105	28	104	33.0%	1.67 [0.93, 3.00]	2016
Kantarjian 2017	91	267	57	109	35.7%	0.47 [0.30, 0.74]	2017
Kantarjian 2019	20	164	20	143	31.3%	0.85 [0.44, 1.66]	2019
Total (95% CI)		**536**		**356**	**100.0%**	**0.86 [0.40, 1.87]**	
Total events	151		105				

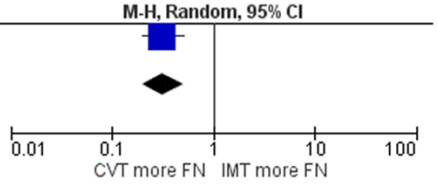
Heterogeneity: Tau² = 0.38; Chi² = 11.30, df = 2 (P = 0.004); I² = 82%
Test for overall effect: Z = 0.38 (P = 0.71)



(f)

Study or Subgroup	IMT Events	IMT Total	CVT Events	CVT Total	Weight	Odds Ratio M-H, Random, 95% CI	Year
Kantarjian 2019	44	164	87	162	100.0%	0.32 [0.20, 0.50]	2019
Total (95% CI)		**164**		**162**	**100.0%**	**0.32 [0.20, 0.50]**	
Total events	44		87				

Heterogeneity: Not applicable
Test for overall effect: Z = 4.87 (P < 0.00001)



Abbreviations: *CR* Complete remission, *CVT* Conventional chemotherapy, *FN* Febrile neutropenia, *IMT* Immunotherapy, *MRD* Minimal residual disease negativity, *OS* Overall survival, *RFS* Relapse-free survival

Supplementary data 4 Forest plot of the subgroup analysis in adult patients: (a) OS rate between immunotherapy into the induction or salvage therapy versus chemotherapy only; (b) RFS rate between immunotherapy into the induction or salvage therapy versus chemotherapy only; (c) CR rate between immunotherapy into the induction or salvage therapy versus chemotherapy only; (d) MRD negativity rate between immunotherapy into the induction or salvage therapy versus chemotherapy only; (e) Infection rate between immunotherapy into the induction or salvage therapy versus chemotherapy only; and (f) Febrile neutropenia rate between immunotherapy into the induction or salvage therapy versus chemotherapy only